

1 Purpose and Motivations

No electrical device is 100% efficient. For instance, a standard household incandescent light bulb converts roughly 2% of the input power into light; the remaining 98% of the power (typically 60–100 W) emerges locally as heat. Few devices are more than 10–20% efficient at converting input power into a form other than heat.

When this heat is an undesirable consequence of operating an electrical device, we must be concerned with how that heat is managed. Excessive heating may adversely affect the performance of the electrical device in question and other nearby systems, or may pose a safety risk.

Radiative versus convective heat transfer.

2 Fundamentals

2.1 Mass and Momentum Conservation

2.2 Energy Conservation

Mechanical energy; internal energy (thermal, elastic, chemical, etc.); first law; meaning of heat and heat and exchange; first law recast as equation for temperature; compressibility; specific heats and relation between them; specific heats for incompressible fluid

$$dq = dh - V dp \quad (1)$$

$$dq = dh - V dp = c_p dT - V dp = c_p dT \left(1 - \frac{V}{c_p} \frac{dp}{dT} \right) \quad (2)$$

2.3 Dimensionless Numbers

2.4 Boundary Conditions

3 Boundary Layers Equations

Entry flow problems: thermal and kinematic

4 Convective Heat Transfer

4.1 Planar Interface

4.2 Rectangular Channel

4.3 Circular Channel

4.4 Applications and Numbers

Air; Water; Aircraft; Submarine; Fan/Pump Power; Emphasize basic scalings; This is the prototype example that exhibits various scalings and sensitivities

5 Fins

The motivation for extending fins into the flow—in effect, cutting channels into the heated surface—is now clear: we are increasing the area of the heated surface in contact with the flow. It is also clear from our earlier discussions that there are constraints on the geometry of the fins (width, height, spacing, and length) that are set by the physical properties of the fluid and substrate, as well as the total power that needs to be dissipated.

6 Optimum Scaling

[Do the scaling problem and identify dimensionless coefficients. One of our goals will be to calculate these coefficients in the context of a basic theory.]

Tuckerman & Pease (1981) derived the basic scaling relations for microchannel cooling. The essence of the idea is that fluid flows through narrow channels that cut across a heated surface. Thermally conducting material into which the channels are cut acts to distribute the heat over a surface area that is larger than the cross section of the heated substrate. A greater heated area in contact with the working fluid increases the rate at which the fluid accepts heat and carries it away. Suppose the heated substrate has width W and length L . A channel of width w and height h has surface area $L(2h + w)$. If we assume that the space between channels is also w , then the total number of channels is approximately $N_c = W/(2w)$ when N_c is large. Therefore, the total area of contact between the fluid and the heated channels is

$$N_c L(2h + w) = LW \left(\frac{1}{2} + \frac{h}{w} \right). \quad (3)$$

For a fixed h , which, as we will see, is determined by the conductivity of the material, we increase the contact area by decreasing w . Thus, a large number of narrow channels should provide for much more effective cooling than other possible channel geometries. In order to assess the full effectiveness of this cooling scheme, we must better understand the thermal and dynamical properties of the working fluid.

Our approach in deriving the optimized system is similar to that in Tuckerman & Pease (1981), but we adopt a somewhat more physically intuitive starting point. Suppose that the fluid flow is stationary and that the velocity field is invariant along the length of the channel. The kinematic entry flow problem and the impact of this region on the effectiveness of the system will be discussed later. We further assume that the heated material in contact with the fluid has an infinite conductivity, so that the temperature within the material is constant **[Only true when no conducting fluid (with finite conductivity) is present. What is the constant of this problem, even when the substrate has infinite conductivity? It's**

not temperature and it's not the flux field. Integrated flux?]. In general, we must solve the full convection diffusion equation in order to determine the fluid temperature in the channel. However, important insights can be gained by first appealing to some basic physical scaling relations.

If the fluid were stationary with an initial temperature T_0 throughout its bulk and temperature $T_w > T_0$ at the channel walls, heat diffuses toward the center of the channel. The time t_w for the heat to diffuse over the half-width of the channel w is given by

$$t_w \sim \frac{w^2}{\alpha} , \quad (4)$$

where α is the thermal diffusivity of the fluid ($\sim 0.01 \text{ cm}^2 \text{ s}^{-1}$ for liquid water). Now, suppose that the fluid flows with a uniform velocity of magnitude u_0 ; this “plug-flow” approximation is reasonable sufficiently near the inlet and away from the walls. In a frame moving with the fluid at speed u_0 the flow appears stationary. The heat diffusion time in eq. (4) and the distance traveled by a fluid element are now related by $t_w = z_e/u_0$, so that the *thermal entry length* z_e is

$$z_e \sim \frac{w^2 u_0}{\alpha} . \quad (5)$$

At this point it is useful to introduce three common dimensionless variables associated with the flow and its thermal transport. A fluid with thermal diffusivity α and kinematic viscosity ν has a Prandtl number

$$\text{Pr} = \frac{\nu}{\alpha} . \quad (6)$$

The length w is characteristic of the flow when the velocity profile is significantly developed beyond the plug flow limit. In this case, the effective Reynolds number is

$$\text{Re} = \frac{u_0 w}{\nu} . \quad (7)$$

With these two numbers in hand, the thermal entry length can be written as

$$z_e \sim w \text{Re} \text{Pr} \equiv w \text{Pe} , \quad (8)$$

where we have introduced the Peclet number Pe . Several other dimensionless numbers will be introduced as needed throughout the discussion.

Once the flow has proceeded beyond the thermal entry length it is well on its way toward thermal equilibrium with the channel walls. Consequently, the heat flux into the fluid decreases and eventually becomes negligible. In some sense, the entry length z_e is the effective “gain region,” where the fluid is capable of accepting a significant flux. Channels longer than z_e offer no additional gain. Therefore, we should construct a microchannel cooling system such that the length of the heated surface along which the channels run is of order z_e .

As before, suppose the heated surface has total width W and a length L of order the thermal entry length z_e . A large number of channels of width w and height $h \gg w$ carry fluid of density ρ at a characteristic speed u_0 . The total area of contact A_c between the fluid and the heated channel walls is approximately (eq. [3])

$$A_c \sim 2W z_e \frac{h}{w} . \quad (9)$$

A single channel carries mass a rate $\dot{m} \sim w h \rho u_0$. If $\Delta T = T_w - T_0$ is the temperature difference between the channel walls and the fluid at the inlet, then $c_p \Delta T$ is the heat per unit mass acquired by the fluid over the channel length. Therefore, the rate at which heat is carried out of the end of the channel, and thus removed from the heated surface, is

$$P_c \sim \dot{m} c_p \Delta T = w h \rho u_0 c_p \Delta T , \quad (10)$$

and the total heating power carried by the N_c channels is $P_{\text{tot}} = N_c P_c$. In steady state, P_{tot} is the total heating power generated by the electronic device and carried to the heated surface. We now find the

fundamental relationship between the heat flux from the substrate $q = P_{\text{tot}}/LW$ and the properties of the fluid and the channel geometry that are required in order to carry off that flux:

$$q \sim \frac{hu_0\rho c_p\Delta T}{z_e} . \quad (11)$$

We can simplify this relation by recalling eq. (5) and noting that $\alpha = \kappa_f/\rho c_p$, where κ_f is the thermal conductivity of the fluid:

$$q \sim \frac{h\kappa_f\Delta T}{w^2} . \quad (12)$$

It would seem that the capacity of the channeled cooling system to carry heat increases indefinitely with the channel aspect ratio h/w . Are there independent constraints on h and w that will allow us to completely specify the geometric, kinematic, and thermal properties of the system? In fact, there are two independent constraints that we can impose, as we show in what follows.

So far, we have neglected the thermal conductivity κ_w of the material making up the channel walls. In effect, we have assumed that κ_w is infinite. Consider a fixed z coordinate along a given fin between fluid channels. Suppose that in the given x - y plane, the fluid is at a temperature $T_f(z)$. At the base of the fin, the temperature takes its maximum value $T_w(z)$. Steady conduction in the fin implies that the supplied flux q flows down a temperature gradient of magnitude

$$\left| \frac{dT}{dy} \right| \sim \frac{q}{\kappa_w} , \quad (13)$$

where $|dT/dy| \sim \Delta T/h$, and thus

$$h \sim \frac{\kappa_w\Delta T}{q} . \quad (14)$$

Clearly, the fluid temperature changes along the channel, and so h above is a maximum at the channel inlet, where $T_f = T_0$. Substituting for h in eq. (12), we obtain

$$q \sim \frac{(\kappa_f\kappa_w)^{1/2}\Delta T}{w} . \quad (15)$$

Moreover, we see that

$$\frac{h}{w} \sim \left(\frac{\kappa_w}{\kappa_f} \right)^{1/2} . \quad (16)$$

7 Conductive Heating in a Constant Flow Field

8 Laminar Channel Flow

Consider the stationary flow of a liquid between two semi-infinite parallel plates, as depicted in Fig. 1 and described in the caption. We assume that the fluid enters the channel at $z = 0$ with a uniform velocity $u_0\hat{z}$. The no-slip condition at the channel walls causes a boundary layer to develop at each wall. As z increases down the length of the channel, the width of each boundary layer increases. Asymptotically, the two boundary layers meet at the center of the channel and the flow profile no longer changes with z ; we say that the flow is now *fully developed*. Before addressing the problem of boundary layer development—the so-called *entry flow problem*—we will first calculate the velocity field for fully developed laminar flow.

8.1 Fully Developed Flow

The dynamics of a viscous, incompressible fluid are governed by the Navier-Stokes equation,

$$\rho \frac{d\mathbf{u}}{dt} = -\nabla p + \mu \nabla^2 \mathbf{u} , \quad (17)$$

and the mass continuity equation

$$\nabla \cdot \mathbf{u} = 0 , \quad (18)$$

where ρ is the mass density, p is the pressure, μ is the dynamic viscosity, and d/dt is the Lagrangian derivative. Here we are neglecting any thermal transport. In the two-dimensional flow being considered, the fluid velocity is $\mathbf{u} = u_x \hat{\mathbf{x}} + u_z \hat{\mathbf{z}}$. Since, by assumption, there is no continued evolution of the velocity field with z in the fully developed flow, we must have $u_x = 0$. If $u_x \neq 0$, then there would be convergence or divergence of the flow, which would cause u_z to depend on both x and z in order to preserve $\nabla \cdot \mathbf{u} = 0$. It follows that in the fully developed flow $u_x = 0$ and $u_z = u_z(x)$. With these simplifications and the assumption of stationarity ($\partial_t \mathbf{u} = 0$), the components of the Navier-Stokes equation reduce to

$$\begin{aligned} 0 &= -\partial_x p , \\ 0 &= -\partial_z p + \mu \partial_x^2 u_z . \end{aligned} \quad (19)$$

We now see that both p and $\partial_z p$ are both functions of only z . Because $\partial_x^2 u_z$ depends on only x and $\partial_z p$ is a function of only z , it must be that these terms are equal to the same constant, which we denote by $-C$. For a stationary flow in the $+z$ direction in which the viscous forces in the $-z$ direction are balanced by the pressure forces ($\propto -\nabla p$) in the $+z$ direction, we have $-\partial_z p = C > 0$. The solution of $\mu \partial_x^2 u_z = -C$ is a quadratic in x that satisfies the no-slip boundary conditions at $x = -w$ and $x = w$:

$$u_z(x) = \frac{Cw^2}{2\mu} \left(1 - \frac{x^2}{w^2} \right) . \quad (20)$$

In a stationary flow, the total rate $\dot{M}$ at which mass crosses a surface that cuts across the channel is a constant equal to

$$\dot{M} = \int d\mathbf{A} \cdot (\rho \mathbf{u}) . \quad (21)$$

For the fully developed flow problem considered here, mass conservation implies

$$\int_{-w}^w dx u_z(x) = \text{constant} . \quad (22)$$

If we set the constant based on the inlet condition ($u_z(x) = u_0$), we have

$$\frac{Cw^2}{3\mu} = u_0 . \quad (23)$$

This relationship allows us to set the pressure gradient based on the inlet speed and the channel width. We can now write for the fully developed velocity profile

$$u_z(x) = \frac{3}{2} u_0 \left(1 - \frac{x^2}{w^2} \right) , \quad (24)$$

where we see that the central speed is $3u_0/2$.

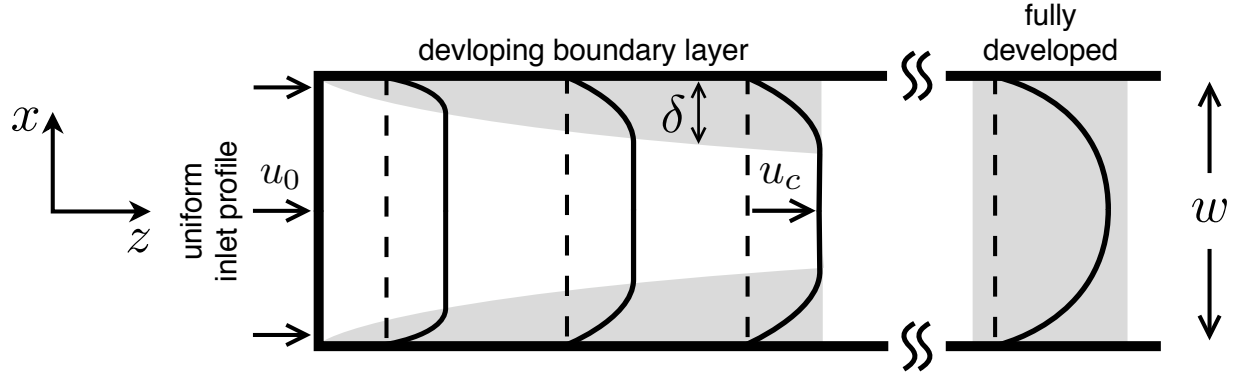


Figure 1: A liquid flows between two semi-infinite parallel plates separated by a width $2w$. The direction of net flow is along the z axis, the x axis is perpendicular the plates, and the y axis points in and out of the page. At the inlet of the channel ($z = 0$), the fluid velocity is assumed to be in the z direction, independent of x , and of magnitude u_0 . As the developing boundary layers (shown as the grey regions) of width $\delta(z)$ converge on the center of the channel, the central speed $u_c(z)$ must increase in order for the net volumetric flow rate to be conserved. In the limit $z \rightarrow \infty$, the boundary layers meet at the center and the velocity field has only a $\hat{z}$ component and is independent of z (i.e., $\mathbf{u} = u_z(x)\hat{z}$). The solid curves crossing the channel show schematically the flow profile $u_z(x)$.

8.2 Entry Flow Problem

The boundary layer grows along the channel as fluid friction transmits information about the no-slip condition at the wall to the fluid closer to the center of the channel. If we neglect the pressure gradient in eq. (17), the cross-channel transport of the z component of fluid momentum roughly follows from

$$u_z \partial_z u_z \sim \nu \partial_x^2 u_z, \quad (25)$$

where we have written $\nu \equiv \mu/\rho$, the kinematic viscosity. As we move from the wall toward the center of the channel at a given z , the fluid speed varies from 0 to roughly u_0 over a distance $\delta(z)$, the characteristic width of the boundary layer. Dimensional analysis of eq. (25) indicates that

$$\delta^2(z) \sim \frac{\nu z}{u_0}. \quad (26)$$

This is essentially the scaling for a diffusing quantity in a static medium, where the mean-square statistical spread after an elapsed time t is $\delta^2 \sim Dt$ for some diffusion coefficient D . In the present problem, $D = \nu$ and the elapsed time is $t = z/u_z$, the time for a fluid element at the center of the channel to move a distance z . Note that

$$\frac{\delta^2}{w^2} \sim \frac{\nu z}{u_0 w^2}. \quad (27)$$

If we define the effective Reynolds number by $\text{Re} = u_0 w / \nu$, we have

$$\frac{\delta^2}{w^2} \sim \frac{z}{\text{Re} w}. \quad (28)$$

As $\delta \rightarrow w$, the two boundary layers meet and the flow becomes fully developed; this occurs over a length $\sim \text{Re} w$. For a given x coordinate not too close to the walls, the length $l \equiv \text{Re} w$ is the order-of-magnitude scale in the z direction over which fluid quantities are expected to change significantly. Very close to the walls, variations in fluid quantities may be appreciable over lengths $z \ll l$.

In the developing flow, u_z depends on both x and z , rather than just x as in the fully developed profile. Fluid near the wall slows down ($\partial_z u_z < 0$), while fluid near the center of the channel must speed up

($\partial_z u_z > 0$) in order to preserve a constant net mass flow rate. It follows that $u_x \neq 0$, since, from the continuity equation, $\partial_x u_x = -\partial_z u_z \neq 0$. We can make intuitive sense of this as follows. Note that $\partial_z u_z \neq 0$ implies a relative acceleration of nearby fluid elements along the z direction, which would cause the elements to move closer or farther apart for $\partial_z u_z < 0$ or > 0 , respectively. For instance, if two nearby fluid elements with the same x coordinate move farther apart, the tendency is for the local density to decrease. Since we are assuming that the fluid is incompressible, fluid in the neighborhood of the two elements in question must move in the x direction (i.e., $u_x \neq 0$) to effectively fill the extra space. A streamline that begins at coordinate $(x = x_0, z = 0)$ at the inlet must curve somewhat toward the channel center as the edge of the boundary layer progressing from the nearest wall crosses x_0 .

Our arguments above regarding the length along the channel over which the boundary layer develops lead us to expect that

$$\partial_z u_z \sim \frac{u_0}{\text{Re } w} . \quad (29)$$

Once the boundary layer thickness is a significant fraction of $w/2$, we expect that the maximum variation of $|u_x|$ in the x direction occurs over a scale of $\lesssim w/2$. So, let $\tilde{u}_x$ be the net variation of $|u_x|$ across the channel at a given z , and write

$$\partial_x u_x \gtrsim \frac{\tilde{u}_x}{w} . \quad (30)$$

Using the continuity equation, we now infer that

$$\tilde{u}_x \lesssim \frac{u_0}{\text{Re}} . \quad (31)$$

Now that the length and velocity scales are in hand, what can we say about the variation of pressure in the channel? Near the center of the channel, the flow is nearly inviscid, in which case the pressure and inertial forces are roughly in balance:

$$\rho |u_z \partial_z u_z| \sim |\partial_z p| , \quad (32)$$

and thus p is expected to vary by of order ρu_0^2 .

Now that we understand the relevant length and velocity scales in the entry flow problem, we can construct a suitable dimensionless form of the Navier-Stokes equation that highlights the importance of the various terms. To this end, we define the following dimensionless variables:

$$x' = \frac{x}{w} , \quad z' = \frac{z}{\text{Re } w} , \quad u'_x = \frac{u_x}{u_0} \text{Re} , \quad u'_z = \frac{u_z}{u_0} , \quad p' = \frac{p}{\rho u_0^2} . \quad (33)$$

The x and z components of the Navier-Stokes equation (with $\partial_t \mathbf{u} = 0$) become

$$\begin{aligned} (u'_x \partial_{x'} + u'_z \partial_{z'}) u'_x &= -\text{Re}^2 \partial_{x'} p' + \left(\partial_{x'}^2 + \frac{1}{\text{Re}^2} \partial_{z'}^2 \right) u'_x , \\ (u'_x \partial_{x'} + u'_z \partial_{z'}) u'_z &= -\partial_{z'} p' + \left(\partial_{x'}^2 + \frac{1}{\text{Re}^2} \partial_{z'}^2 \right) u'_z . \end{aligned} \quad (34)$$

The transformed divergence-free condition contains no explicit dependence on Re :

$$\partial_{x'} u'_x + \partial_{z'} u'_z = 0 . \quad (35)$$

Following the classical discussions of boundary layer theory, we make the simplifying assumption that $\text{Re} \gg 1$, which turns out to be a rather robust assumptions over most of the channel. For simplicity in notation, we drop the primes that denote the dimensionless variables, and we have in the $\text{Re} \rightarrow \infty$ limit,

$$0 = \partial_x p , \quad (36)$$

$$(u_x \partial_x + u_z \partial_z) u_z = -\partial_z p + \partial_x^2 u_z , \quad (37)$$

$$\partial_x u_x + \partial_z u_z = 0 . \quad (38)$$

Just as in the fully developed flow, we see that $p = p(z)$.

The solution of the reduced equations must satisfy several constraints. We know that the flow must asymptotically approach the fully developed solution discussed in § 8. Moreover, the solution must satisfy the conservation of net mass and momentum flow rate. Integral forms of mass and momentum conservation are easily obtained by integrating eqs. (37) and (38) over the width of the channel. For mass continuity, we have

$$0 = \int_{-1}^1 dx (\partial_x u_x + \partial_z u_z) = u_x(1, z) - u_x(-1, z) + \partial_z \int_{-1}^1 dx u_z(x, z) . \quad (39)$$

The two boundary terms vanish due to the no-slip condition, and so we have, just as in eq. (22),

$$\int_{-1}^1 dx u_z(x, z) = 2 , \quad (40)$$

where the constant on the right-hand side has been set by the inlet condition $u_z(x, 0) = 1$. Integrating the momentum equation, we obtain

$$\begin{aligned} 0 &= \int_{-1}^1 dx [u_x \partial_x u_z + u_z \partial_z u_z + \partial_z p - \partial_x^2 u_z] \\ &= \int_{-1}^1 dx \left[\partial_x (u_x u_z) - u_z \partial_x u_x + \frac{1}{2} \partial_z u_z^2 + \partial_z p - \partial_x^2 u_z \right] . \end{aligned} \quad (41)$$

Replacing $\partial_x u_x$ with $-\partial_z u_z$, integrating the boundary terms, and using the no-slip condition where appropriate, we have

$$\frac{d}{dz} \left[\int_{-1}^1 dx u_z^2 \right] = -2 \partial_z p + \partial_x u_z(1, z) - \partial_x u_z(-1, z) . \quad (42)$$

Because of the symmetry of the flow profile, we know that $\partial_x u_z(-1, z) = -\partial_x u_z(1, z)$, and so the momentum integral is

$$\frac{d}{dz} \left[\frac{1}{2} \int_{-1}^1 dx u_z^2 \right] = -\partial_z p + \partial_x u_z(1, z) . \quad (43)$$

A complete, analytic solution to eqs. (36)-(38) does not exist. Here we use the physically motivated, but otherwise ad hoc, approach of Fargie & Martin (1971), who considered the developing laminar flow in circular pipes. Their results compare favorably to other theoretical results and experimental measurements of viscous pipe flow. We expect that our adaptation of the ideas in Fargie & Martin (1971) to planar flow will prove to be a suitable approximation for our purposes.

The starting point in the analysis of Fargie & Martin (1971) is to assume a simple functional form for $u_z(x, z)$. As the boundary layers develop from the walls, the fluid near the center of the channel ($|x| < 1 - \delta$) should have a nearly flat velocity profile across the channel for a given z . Moreover, we know that in the fully developed flow, $u_z(x)$ is parabolic. For these reasons, Fargie & Martin (1971) adopted an evolving velocity profile that is parabolic in the boundary layers and flat in the channel “core.” We follow the same prescription and set $u_z(x, z)$ equal to the following piecewise continuous function, where $\alpha(z) = 1 - \delta(z)$:

$$u_z(x, z) = \begin{cases} u_c \left[1 - \left(\frac{|x| - \alpha}{1 - \alpha} \right)^2 \right] , & \alpha < |x| < 1 \\ u_c , & |x| < \alpha \end{cases} \quad (44)$$

This function satisfies the symmetry of the problem as well as the no-slip condition at the walls. Our next task is to determine the core speed $u_c(z)$ and core width $\alpha(z)$ in a way that is consistent with integral mass and momentum conservation.

Mass conservation (eq. [40]) yields the simple result,

$$u_c = \frac{3}{2 + \alpha} , \quad (45)$$

which has the correct limit of $3/2$ as $\alpha \rightarrow 0$ when the flow is fully developed (see eq. [24]). For the momentum integral (eq. [43]), we have the pieces

$$\frac{1}{2} \int_{-1}^1 dx u_z^2 = \int_0^1 dx u_z^2 = \frac{u_c^2}{15} (8 + 7\alpha) = \frac{3(8 + 7\alpha)}{5(2 + \alpha)^2} , \quad (46)$$

and

$$\partial_x u_z(1, z) = -\frac{2u_c}{1 - \alpha} = -\frac{6}{(1 - \alpha)(2 + \alpha)} . \quad (47)$$

But what do we substitute for $\partial_z p$? Although the approximate equations we derived the entry flow are suspect near the walls and the inlet, where the assumption of $\text{Re} \gg 1$ breaks down, a plausible trial solution would be to assume that $\partial_x p = 0$ is exact right up to the wall. Writing eq. (37) at $x = 1$, where $\mathbf{u} = 0$, we find

$$\partial_z p = \partial_x^2 u_z(1, z) = -\frac{2u_c}{(1 - \alpha)^2} = -\frac{6}{(1 - \alpha)^2(2 + \alpha)} . \quad (48)$$

This has the correct limit for fully developed flow ($\alpha \rightarrow 0$), namely $-\partial_z p = 3$ (eq. [23]). However, this expression blows up as $\alpha \rightarrow 1$, because both $|\partial_x u_z|$ and $|\partial_x^2 u_z|$ are formally infinite at the channel inlet. We will explore this issue in more detail after we have presented the main result of this section.

With the transverse pressure gradient given by eq. (48), the momentum integral now gives

$$\frac{d}{dz} \left[\frac{3(8 + 7\alpha)}{5(2 + \alpha)^2} \right] = -\partial_z p + \partial_x u_z(1, z) = \frac{6\alpha}{(1 - \alpha)^2(2 + \alpha)} . \quad (49)$$

After evaluating the derivative, the differential equation for $\alpha(z)$ is

$$\frac{d\alpha}{dz} = -10 \frac{\alpha(2 + \alpha)^2}{(2 + 7\alpha)(1 - \alpha)^2} . \quad (50)$$

Note that as $\alpha \rightarrow 1$ at the inlet, the above equation approaches

$$\frac{d\alpha}{dz} = -\frac{10}{(1 - \alpha)^2} , \quad (51)$$

or

$$\frac{d\delta}{dz} = \frac{10}{\delta^2} , \quad (52)$$

with the solution $\delta(z) = (30z)^{1/3}$. This limiting behavior does not conform with the expected diffusion scaling $\delta \propto z^{1/2}$ (eq. [26]). However, since we have a certain amount of freedom in how we construct our ad hoc solution, we can recover the diffusion scaling near the inlet by replacing $(1 - \alpha)^2$ in eq. (50) with $(1 - \alpha)$. This is equivalent to assuming

$$-\partial_z p = \frac{6(1 + \alpha)}{(1 - \alpha)(2 + \alpha)} , \quad (53)$$

a function that preserves the correct limiting behavior of $\partial_z p$. Let us now write

$$\frac{d\alpha}{dz} = -10 \frac{\alpha(2 + \alpha)^2}{(2 + 7\alpha)(1 - \alpha)} . \quad (54)$$

In the limit of $\alpha \rightarrow 0$, the above equation reduces to

$$\frac{d\alpha}{dz} = -20\alpha , \quad (55)$$

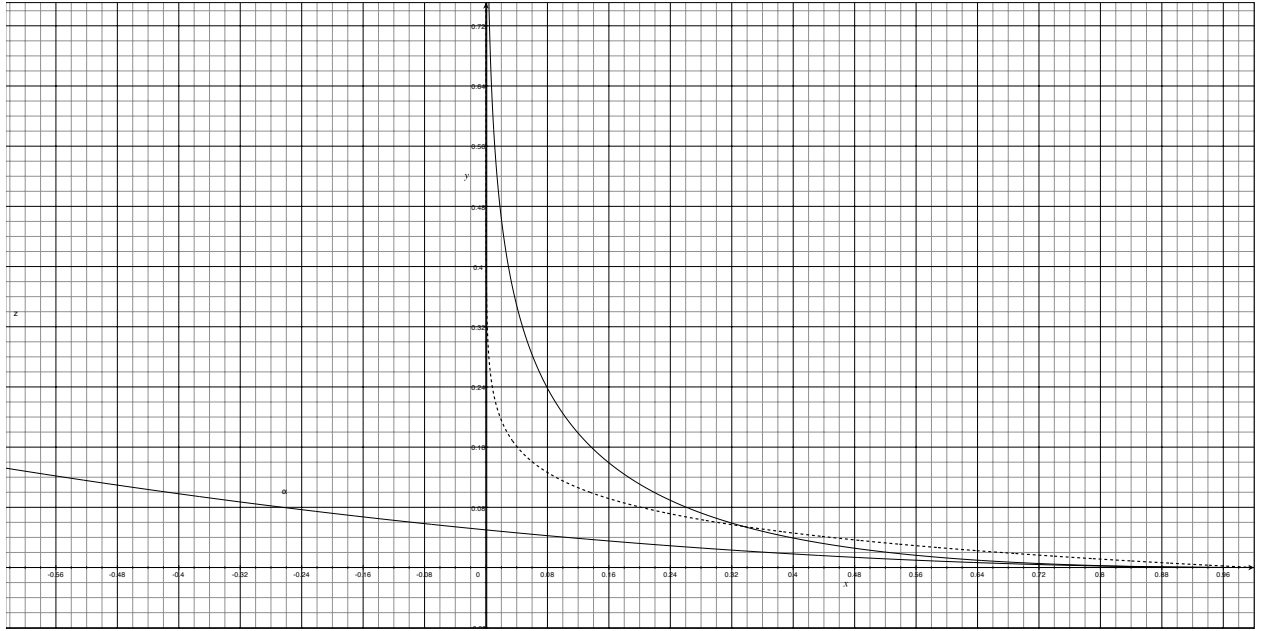


Figure 2: Curves show $z(\alpha)$ for the complete solution to eq. (54) (solid) and the exponential function eq. (55) for the $\alpha \rightarrow 0$ limit (dashed).

with the solution

$$\alpha = e^{-20z} . \quad (56)$$

Based on this limiting relation, we expect that the full solution will show that the core begins to narrow quickly along the channel while z is substantially less than 1. Indeed, the complete solution to eq. (54) does show this behavior. Although the complete solution can be obtained analytically, the result is not very illuminating, and so we display the result in graphical form in Fig. (2), where we have overlayed the limiting solutions for $\alpha \rightarrow 1$ and $\alpha \rightarrow 0$.

Reading from Fig. (2), we see that $z \simeq 0.2$ when $\alpha = 0.1$ and $z \simeq 0.5$ when $\alpha = 0.01$. From these numbers, we can define the corresponding entry lengths as $z_{90} \simeq 0.2$ and $z_{99} \simeq 0.6$ for $1 - \alpha = 0.9$ and 0.99 , respectively. As a crude estimate, we surmise that the *entry length* along the planar channel, the distance over which the fully developed flow is nearly attained, is $z_e \sim 0.5$. In dimensional length units, the entry length is of order $\text{Re } w$, as expected. Note that in a circular pipe of radius w , the entry length is closer to $\text{Re } w/20$, or roughly an order of magnitude smaller than in the planar case. **[I really need to check this stuff; this seems fishy.]**

9 Heat Transfer Along a Channel

In this section, we address the problem of heat transfer from a solid material at a fixed temperature to a fluid moving through a channel of some cross section cut through the material. We assume that the situation is stationary and that the fluid heat capacity and thermal conductivity are constant. Moreover, as per our earlier discussions (**fill in section refs**), we neglect heat conduction along the length of the channel, viscous heating, and the components of the fluid velocity perpendicular to the channel axis. It follows that the dimensionless equation to be solved is

$$u_z \partial_z \theta = \nabla_{\perp}^2 \theta , \quad (57)$$

where θ is a linear function of the temperature, and $\nabla_{\perp}^2$ is the two-dimensional Laplacian in the plane perpendicular to the channel axis.

What is a suitable definition for θ ? There are two fixed temperatures to consider, the initial fluid temperature T_0 that the fluid has upon entering the channel and the fixed temperature T_w of channel walls. At the channel walls the fluid temperature T must equal T_w . It is convenient to define θ such that $\theta = 0$ at the walls. Therefore, we write

$$\theta = \frac{T - T_w}{T_0 - T_w} . \quad (58)$$

The boundary conditions needed to solve eq. (57) are $\theta = 1$ at the inlet and $\theta = 0$ at the walls. Also, as z increases, the temperature profile across the channel flattens and approaches the wall temperature as $z \rightarrow \infty$; in other words, $\theta(x, z \rightarrow \infty) = 0$.

In what follows, we consider two different geometries, (1) a circular pipe with a radius equal to 1 in dimensionless units, and (2) a plane parallel channel that is infinite in the y dimension with walls at $x = -1$ and $x = 1$. For the planar channel, we simply have $\nabla_{\perp}^2 = \partial_x^2$, while for the circular pipe,

$$\nabla_{\perp}^2 = r^{-1} \partial_r (r \partial_r) = \partial_r^2 + r^{-1} \partial_r , \quad (59)$$

where r is the radius measured from the central axis. We can combine these results by using x to represent the cross channel dimension in both cases and writing

$$\nabla_{\perp}^2 = \partial_x^2 + \frac{s}{x} \partial_x , \quad (60)$$

where $s = 0$ or 1 for the planar or cylindrical geometries respectively.

If we restrict our attention to flow that is fully developed kinematically (**refer to justification**), then we know that u_z is a function of only x . When we rewrite eq. (57) as

$$\partial_z \theta = \frac{1}{u_z(x)} \left(\partial_x^2 + \frac{s}{x} \partial_x \right) \theta , \quad (61)$$

it is obvious that we should seek a separable trial solution of the form $\theta = F(x)G(z)$; this approach yields

$$\frac{1}{G} \frac{dG}{dz} = \frac{1}{u_z(x)F(x)} \left(\frac{d^2 F}{dx^2} + \frac{s}{x} \frac{dF}{dx} \right) . \quad (62)$$

Since the left side is a function of only z and the right side is a function of only x , then both sides must be equal to the same numerical constant. If this constant is a negative number, then $G(z)$ decreases exponentially along the channel and $\theta(x, z \rightarrow \infty) = 0$, as required. Therefore, we set both sides of eq. (62) equal to $-k^2$ and find

$$G(z) \propto e^{-k^2 z} \quad (63)$$

and

$$\frac{d^2 F}{dx^2} + \frac{s}{x} \frac{dF}{dx} + k^2 u_z(x) F = 0 . \quad (64)$$

We are now left with the task of solving an ordinary second-order differential equation for F subject to the appropriate boundary conditions. In general, the boundary conditions will be satisfied only for certain values of k , called the *eigenvalues* of the problem. For each eigenvalue k_n in the set $\{k_0, k_1, \dots\}$ there is a solution *eigenvector* $F(x; k_n)$, for which we use the shorthand $F_n(x)$. Similarly, we write $G_n(z) \equiv G(z; k_n)$. The general solution for θ is a sum over the linearly independent eigenvectors $F_n(x)G_n(z)$:

$$\theta(x, z) = \sum_{n=0} a_n F_n(x) G_n(z) , \quad (65)$$

where the a_n are constant coefficients to be determined by one of the boundary conditions on θ .

Interject orthogonality of eigenvalues above somehow. Linear independence and orthog?

[Better introduce general eigenvalue problem and write symbolic series expansion for θ . What is the general description of a boundary value problem? Carry on with this analysis; show orthogonality; derive formal solution for series coefficients; apply to plug, then cylindrical (solution known, Nusselt?), then planar.]

9.1 Plug Flow

Before taking on eq. (64) with the parabolic velocity profile of fully developed channel flow, we consider the instructive and simpler example of “plug” flow, where fluid viscosity is neglected and $u_z(x, z) = 1$ in our dimensionless units. Furthermore, let us focus on planar channel flow with $s = 0$. The equation for F is now

$$\frac{d^2 F}{dx^2} + k^2 F = 0 , \quad (66)$$

with the well known general solution

$$F(x; k) = A \sin(kx) + B \cos(kx) . \quad (67)$$

Because of the inherent symmetry of the problem across the $x = 0$ plane, we see that $A = 0$. In order to satisfy the boundary condition that $\theta = 0$ at the walls, we must demand that $F(x = \pm 1; k) = 0$; this holds when

$$k = (2n + 1) \frac{\pi}{2} \equiv k_n \quad (68)$$

for integers $n \geq 0$. It follows that for plug flow, the temperature profile in the channel is

$$\theta = \sum_{n=0}^{\infty} a_n e^{-k_n^2 z} \cos(k_n x) . \quad (69)$$

The coefficients a_n are determined from the inlet boundary condition $\theta(x, z = 0) = 1$:

$$1 = \sum_{n=0}^{\infty} a_n \cos(k_n x) \quad (70)$$

Multiply both sides by $\cos(k_m x)$ and integrate:

$$\int_{-1}^1 dx \cos(k_m x) = \sum_{n=0}^{\infty} a_n \int_{-1}^1 dx \cos(k_m x) \cos(k_n x) . \quad (71)$$

It is easy to show that

$$\int_{-1}^1 dx \cos(k_m x) \cos(k_n x) = \delta_{mn} , \quad (72)$$

and thus

$$a_n = \frac{2 \sin k_n}{k_n} = \frac{4(-1)^n}{(2n + 1)\pi} . \quad (73)$$

The full solution for the temperature profile is now

$$\theta(x, z) = \frac{4}{\pi} \left[e^{-\pi^2 z/4} \cos(\pi x/2) - \frac{1}{3} e^{-9\pi^2 z/4} \cos(3\pi x/2) + \dots \right] . \quad (74)$$

Consider the ratio

$$\frac{e^{-k_1^2 z}}{e^{-k_0^2 z}} = e^{-2\pi^2 z} . \quad (75)$$

When $z > (2\pi^2)^{-1} \simeq 1/20$, the first term in eq. (74) is more than $3e$ times larger than the second term; in this regime we have the excellent approximation

$$\theta(x, z) = \frac{4}{\pi} e^{-\pi^2 z/4} \cos(\pi x/2) . \quad (76)$$

10 Heat Transfer in the Channel [Deprecated]

Now that we grasp the dynamics of the flow between parallel plates, we can now consider the coupling between the fluid flow and the heat transfer in the channel. For simplicity, we solve the heat transfer problem for an assumed flow field. That is, we use the results of the last section for $u_z(x, z)$ and solve the heat transfer equation for the fluid temperature $T(x, z)$ in the channel. The evolution of the temperature in the channel is governed by the equation

$$\rho c_p \frac{dT}{dt} = \kappa_f (\partial_x^2 + \partial_z^2) T + \mu (\partial_x u_z)^2, \quad (77)$$

where μ is the fluid's dynamic viscosity. We assume here that μ and the thermal conductivity κ_f are independent of temperature. In the same spirit as § 8.2, we derive a dimensionless form of the heat equation to highlight the most important terms.

With the same dimensionless variables in eq. (33), along with the dimensionless temperature

$$\theta = \frac{T - T_0}{T_w - T_0} \equiv \frac{T - T_0}{\Delta T}, \quad (78)$$

the heat equation transforms to

$$\frac{\Delta T u_0}{\text{Pe } w} u'_z \partial_{z'} \theta = \alpha \Delta T \left(\frac{1}{w^2} \partial_{x'}^2 + \frac{1}{\text{Pe}^2 w^2} \partial_{z'}^2 \right) \theta + \frac{\nu u_0^2}{c_p w^2} (\partial_{x'} u'_z)^2. \quad (79)$$

After some rearrangements, we find

$$u'_z \partial_{z'} \theta = \partial_{x'}^2 \theta + \frac{1}{\text{Pe}^2} \partial_{z'}^2 \theta + \text{Pr Ec} (\partial_{x'} u'_z)^2, \quad (80)$$

where Ec is the Eckert number, defined by

$$\text{Ec} = \frac{u_0^2}{c_p \Delta T}, \quad (81)$$

which measures the ratio of the kinetic energy of the flow to its net enthalpy gain.

From previous arguments, it seems reasonable to neglect the term proportional to Pe^{-2} . However, for the time being we will keep the $\mathcal{O}(\text{Pr Ec})$ term. Dropping the primes and letting $\text{Pr Ec} \equiv V$ be the viscous heating coefficient, we have

$$u_z \partial_z \theta = \partial_x^2 \theta + V (\partial_x u_z)^2. \quad (82)$$

We will consider two regimes that allow an analytic solution to eq. (82). In both cases, we assume a given form for $u_z = u_z(x)$; that is, we neglect the development of the flow field along the length of the channel. First, we study plug flow, where $u_z = 1$ in our dimensionless formulation, and we follow this with a similar analysis for fully developed laminar flow with $u_z(x) = \frac{3}{2}(1 - x^2)$ (eq. [24]).

In the plug flow solution, we set $V = 0$, since we are effectively neglecting fluid viscosity. Our task is now to solve the simpler equation

$$\partial_z \theta = \partial_x^2 \theta, \quad (83)$$

subject to the boundary conditions that $\theta = 0$ at the walls ($x = \pm 1$) and $\theta = 1$ at the inlet ($z = 0$) for $-1 < x < 1$. Substituting the separable eigenfunction $\theta(x, z) = F(x)G(z)$, we find

$$\frac{\partial_z G}{G} = \frac{\partial_x^2 F}{F}. \quad (84)$$

Both sides must be equal to same constant, which we set to $-k^2$. The resulting eigenfunction is

$$F(x)G(z) \propto e^{-k^2 z} \cos(kx), \quad (85)$$

where $\cos(kx)$ satisfies the symmetry of the problem. The boundary conditions at the walls are satisfied if

$$k = (2n + 1) \frac{\pi}{2} \equiv k_n \quad (86)$$

for n an integer ≥ 0 . It follows that the general solution for $\theta(x, z)$ is

$$\theta(x, z) = \sum_{n=0}^{\infty} c_n e^{-k_n^2 z} \cos(k_n x) . \quad (87)$$

The coefficients c_n are determined from the inlet condition $\theta(x, z = 0) = 1$, where

$$1 = \sum_{n=0}^{\infty} c_n \cos(k_n x) . \quad (88)$$

Now multiply both sides by $\cos(k_m x)$ and integrate:

$$\int_{-1}^1 dx \cos(k_m x) = \sum_{n=0}^{\infty} c_n \int_{-1}^1 dx \cos(k_m x) \cos(k_n x) . \quad (89)$$

Noting that $\cos(a) \cos(b) = \frac{1}{2} [\cos(a + b) + \cos(a - b)]$ and carrying out the integrations, we have

$$\frac{2}{k_m} \sin k_m = \sum_{n=0}^{\infty} c_n \left[\frac{\sin(k_m + k_n)}{k_m + k_n} + \frac{\sin(k_m - k_n)}{k_m - k_n} \right] . \quad (90)$$

Since $k_m + k_n = \pi(m + n + 1)$ is a multiple of π , $\sin(k_m + k_n) = 0$. However, $k_m - k_n = \pi(m - n)$, and thus

$$\frac{\sin(k_m - k_n)}{k_m - k_n} = \delta_{mn} , \quad (91)$$

so that

$$c_n = \frac{2}{k_n} \sin k_n = \frac{4}{(2n + 1)\pi} (-1)^n . \quad (92)$$

The exponential term in eq. (87) effectively truncates the series at relatively small n , depending on the value of z . For instance, when $z = (2\pi^2)^{-1}$ the ratio $\exp(-k_1^2 z) / \exp(-k_0^2 z)$ equals e^{-1} . When z is in this regime, we have the useful approximation

$$\theta(x, z) \simeq \frac{4}{\pi} e^{-\frac{\pi^2}{4} z} \cos\left(\frac{\pi}{2} x\right) . \quad (93)$$

As z increases beyond $4/\pi^2$, the exponential term drives θ to 0, its value at the walls.

It is interesting to point out that fluid flow plays no important role in this plug flow solution. Because there is no gradient in the flow speed, we are free to write $z = t$ for all x .

References

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